Trigonometric Ratio: Formula Sheet

Basic identities

Pythagorean: $\sin^2\theta + \cos^2\theta = 1$

Secant: $\sec^2\theta - \tan^2\theta = 1$

Cosecant:

$$\operatorname{cosec}^2\theta - \cot^2\theta = 1$$

Fourth powers: $\sin^4 + \cos^4 = 1 - 2\sin^2\cos^2$

Sixth powers: $\sin^6 + \cos^6 = 1 - 3\sin^2\cos^2$

Special:

$$\frac{\sin^4\theta}{a} + \frac{\cos^4\theta}{b} = \frac{1}{a+b} \Rightarrow \sin^2\theta = \frac{a}{a+b}$$

Angles and reduction

Conversion: π rad $= 180°$

Odd n in $n\frac{\pi}{2} \pm \theta$: co-ratio

Negative angle:

$$\sin(-\theta) = -\sin\theta,\ \cos(-\theta) = \cos\theta$$

Sign check:

$$\sin A \pm \cos A = \sqrt{2}\sin(A \pm \frac{\pi}{4})$$

Compound angles

Sine:

$$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$$

Cosine:

$$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$$

Tangent:

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$$

Cot:

$$\cot(A \pm B) = \frac{\cot A\cot B \mp 1}{\cot B \pm \cot A}$$

Difference of squares:

$$\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B$$

Cos form:

$$\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B$$

45°:

$$\tan(\frac{\pi}{4} \pm A) = \frac{1 \pm \tan A}{1 \mp \tan A}$$

Three angles:

$$\tan(A+B+C) = \frac{\sum\tan A - \prod\tan A}{1 - \sum\tan A\tan B}$$

Product and sum (C+D)

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$

$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$

$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$

$$\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$$

$$\sin C - \sin D = 2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$$

$$\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$$

$$\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$$

Multiple angles

$$\sin 2A = 2\sin A\cos A = \frac{2\tan A}{1+\tan^2 A}$$

$$\cos 2A = 2\cos^2 A - 1 = 1 - 2\sin^2 A = \frac{1-\tan^2 A}{1+\tan^2 A}$$

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

Half:

$$\tan A = \frac{1-\cos 2A}{\sin 2A} = \frac{\sin 2A}{1+\cos 2A}$$

$$1 + \cos A = 2\cos^2\frac{A}{2},\ 1 - \cos A = 2\sin^2\frac{A}{2}$$

$$1 \pm \sin A = (\cos\frac{A}{2} \pm \sin\frac{A}{2})^2$$

Modulus:

$$|\sin A \pm \cos A| = \sqrt{1 \pm \sin 2A}$$

$$\sin 3A = 3\sin A - 4\sin^3 A$$

$$\cos 3A = 4\cos^3 A - 3\cos A$$

$$\tan 3A = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$$

Standard values

15°:

$$\sin = \frac{\sqrt{3}-1}{2\sqrt{2}},\ \tan = 2 - \sqrt{3}$$

75°:

$$\sin = \frac{\sqrt{3}+1}{2\sqrt{2}},\ \tan = 2 + \sqrt{3}$$

22.5°: $\tan = \sqrt{2} - 1$; 67.5°: $\tan = \sqrt{2} + 1$

18°: $\sin = \frac{\sqrt{5}-1}{4}$

36°: $\cos = \frac{\sqrt{5}+1}{4}$

Important results

$$\cot A - \tan A = 2\cot 2A$$

$$\cot A + \tan A = 2\operatorname{cosec} 2A$$

$A + B = 45°$: $(1 + \tan A)(1 + \tan B) = 2$

$$\sin\theta \sin(60° - \theta) \sin(60° + \theta) = \frac{1}{4}\sin 3\theta$$

$$\cos\theta \cos(60° - \theta) \cos(60° + \theta) = \frac{1}{4}\cos 3\theta$$

$$\tan\theta \tan(60° - \theta) \tan(60° + \theta) = \tan 3\theta$$

Continued product:

$$\prod_{k=0}^{n-1} \cos 2^k\theta = \frac{\sin 2^n\theta}{2^n \sin\theta}$$

Componendo-dividendo:

$$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Triangle ($A + B + C = \pi$)

$$\sum \sin 2A = 4 \sin A \sin B \sin C$$

$$\sum \cos 2A = -1 - 4 \cos A \cos B \cos C$$

$\sum \sin^2 A = 2 + 2\prod \cos A$; $\sum \cos^2 A = 1 - 2\prod \cos A$

$\sum \tan A = \prod \tan A$; $\sum \cot A \cot B = 1$

$\sum \tan \frac{A}{2} \tan \frac{B}{2} = 1$; $\sum \cot \frac{A}{2} = \prod \cot \frac{A}{2}$

Inequalities: $\sum \cos A \le \frac{3}{2}$, $\sum \sin A \le \frac{3\sqrt{3}}{2}$, $\sum \cot^2 A \ge 1$

Ranges

$$a \sin x + b \cos x \in [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$$

$$|\sin x| + |\cos x| \in [1, \sqrt{2}]$$

$$a^2 \tan^2 x + b^2 \cot^2 x \in [2ab, \infty)$$

$$a^2 \sec^2 x + b^2 \operatorname{cosec}^2 x \in [(a+b)^2, \infty)$$

$a^2 \sin^2 x + b^2 \operatorname{cosec}^2 x$: $[2ab, \infty)$ if $b \le a$, else $[a^2 + b^2, \infty)$

Series

A.P. angles:

$$\sum \sin(\alpha + k\beta) = \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}} \sin\left(\alpha + \frac{(n-1)\beta}{2}\right)$$

Split:

$$\frac{\sin(A-B)}{\cos A \cos B} = \tan A - \tan B$$

Split:

$$\frac{\sin(A-B)}{\sin A \sin B} = \cot B - \cot A$$

Split: $\operatorname{cosec} 2x = \cot x - \cot 2x$, $\tan x = \cot x - 2\cot 2x$